

SENSOR-DERIVED SIGIL AND ABSTRACT PHYSIOLOGICAL VISUALIZATION SYSTEM

Device-Generated Symbolic Representation and Compatibility Mapping

Field

This disclosure relates to systems and methods for abstract visualization of physiological data and, more particularly, to the generation of non-explicit symbolic representations derived from multi-device physiological sensor data for interpretation, communication, and comparative analysis between users.

Overview

In certain embodiments, physiological data captured by one or more wearable or insertable devices is transformed into an abstract, symbolic visual representation, referred to herein as a “sigil.” The sigil is generated using processed metrics derived from device sensors and is designed to convey complex physiological patterns in a non-explicit, non-anatomical, and visually interpretable form.

The sigil system enables users, clinicians, or systems to perceive physiological states, trends, or compatibility relationships without displaying raw numerical values or explicit anatomical imagery.

Data Inputs

Sigil generation may incorporate metrics derived from one or more sensing devices, including but not limited to:

- Structural or geometric measurements
- Pressure, strain, or compliance metrics
- Vascular or flow-related indicators
- Temporal patterns of arousal, activation, or recovery
- Autonomic or cardiovascular context signals
- Thermal, hydration, or impedance-based measures
- Stability, symmetry, proportionality, or variability indices

Data may be normalized, weighted, anonymized, or transformed prior to visualization.

Abstract Symbol Generation

The sigil is generated through one or more computational processes that map processed physiological metrics to visual parameters, including but not limited to:

- Geometric shapes, curves, or radial structures
- Line thickness, continuity, or segmentation
- Symmetry, asymmetry, or proportional relationships
- Orientation, rotation, or spatial balance
- Color, gradient, opacity, or intensity
- Layering, overlap, or modular elements

The resulting symbol is intentionally abstract and does not depict explicit anatomy or direct biometric identifiers.

Dynamic and Temporal Behavior

In certain embodiments, sigils may be:

- Static representations of a baseline or averaged state
- Time-varying or animated representations reflecting change over time
- State-dependent representations corresponding to context (e.g., rest, activity, sleep)

Sigils may update as new sensor data is acquired or as metrics evolve.

Compatibility and Comparative Mapping

In certain embodiments, sigils generated for different users may be compared algorithmically to assess compatibility, alignment, or relational metrics. Compatibility analysis may include:

- Structural or proportional similarity
- Complementary or harmonizing metric relationships
- Temporal alignment or synchrony
- Stability or variance matching

Compatibility results may be visualized through sigil overlap, fusion, resonance patterns, composite symbols, or comparative layouts, without revealing explicit physiological data.

Privacy and Abstraction Advantages

The sigil system provides a privacy-preserving visualization layer by abstracting sensitive physiological information into symbolic form. This enables sharing, comparison, or communication of physiological insights without exposing explicit anatomy, raw sensor values, or clinically sensitive details.

System Integration

The sigil generation system may be implemented locally on a device, on a companion application, or on a remote computing system, and may operate independently or in coordination with multi-node physiological monitoring platforms.

Safety and Interpretation Controls

In certain embodiments, sigil generation includes constraints to prevent misinterpretation, including normalization limits, confidence indicators, context gating, or explanatory overlays. Sigils are intended as interpretive or communicative representations rather than diagnostic outputs unless explicitly configured for clinical use.